# Aniketh Bandlamudi

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## Education

**University of Virginia** — Charlottesville, VA
*B.S. in Computer Science | Minors: Applied Mathematics, Philosophy (planned)* — *Expected May 2030*

- *Relevant Coursework:* APMA 2120 (Multivariable Calculus), CS 2100 (Data Structures & Algorithms 1)
- *Activities:* UVA Solar Car Team (Embedded Software), IEEE UVA Student Chapter (Member)

## Experience

**Founder & Engineer** — Jan. 2026 – Present
*Pragmatic (aniketh.net/pragmatic | github.com/pragmaticco/pragmatic)* — *Remote*

- Building a zero-dependency durable execution runtime in Rust that lets nondeterministic AI agents survive crashes and resume exactly where they left off
- Designed an append-only step journal at 230 ns/append with flat latency through 1M entries, passing a 1000/1000 replay-determinism test suite
- Formalized agent execution and replay with a probabilistic process calculus
- Incorporated as Pragmatic Research, Inc. (Delaware C-corp) in Aug. 2026

**Founder & President** — May 2022 – Present
*OCRadar (ocradar.com)* — *Vienna, VA*

- Founded a nonprofit and led a team of 4 building a computer-vision app that flags suspicious oral lesions
- Published a CNN for oral squamous cell carcinoma trained on 4,293 images: 97.8% precision, 95.3% recall, and 98% test accuracy at 1440p, with a 3D-printed capture attachment (arXiv:2510.16235)
- Moved inference on-device with Core ML and added a confidence threshold that abstains instead of guessing
- Featured in outlets reaching millions: WTOP, Korea Times, Vision Times China, FCPS News, Tech Roast Show

**Engineering Intern** — May 2026
*Alarm.com* — *Tysons, VA*

- Configured and integrated home security devices for hardware QA testing
- Soldered buck converters and embedded boards in the hardware lab
- Built an internal software tool to manage lab devices, streamlining QA/QE-to-device-engineering testing cycles

**Software Lead (FRC) / R&D Captain (FTC)** — 2022 – 2026
*Oakton Robotics* — *Vienna, VA*

- Led software for a 25-member team, owning systems engineering and control software for competition robots
- Oversaw robot design and programming over four seasons; earned finalist, innovation, and creativity awards

## Projects

**Haze** | *Swift, SwiftUI, iOS, watchOS, macOS* — Site | App Store

- Solo-built and shipped a 5-star iPhone, iPad, Mac, and Watch app with widgets, Live Activities, and Siri
- Blends ECMWF, GFS, and ICON models, corrects with NWS station data, and sends background rain alerts

**Meridian** | *Rust, Swift, UniFFI, DuckDB, Claude API* — Download | GitHub

- Built a macOS financial terminal on a 53K-line Rust core bridged via UniFFI to SwiftUI, with 35+ functions
- Streams 2,000 live symbols at 60 fps on under 10% of one CPU core from 9 data sources, with a Claude AI analyst

**Hypnagogia** | *Python, Brian2, TypeScript* — Live | GitHub

- Simulated a whole Drosophila brain (144K neurons, 21.9M synapses) to test if a learned memory replays in sleep
- Scored replay against shuffled-connectome and random-ensemble controls; reported negative results in full

**IllumiSign** | *Hack the Nest 2025 Winner* — Devpost | GitHub

- Won Hack the Nest 2025 with a browser-based communication tool that uses real-time eye tracking to let ALS patients select letters and generate speech without specialized hardware

## Technical Skills

**Languages**: Java, C++, Python, Rust, Swift, TypeScript, HTML/CSS/JavaScript
**Technologies**: PyTorch, Computer Vision, iOS/watchOS Development
**Robotics & Hardware**: WPILib, PathPlanner, Limelight, CTRE Phoenix, SysId, Embedded Systems, KiCad
**Developer Tools**: Git, Xcode, VS Code, Claude Code, Codex, Docker, ffmpeg, COLMAP
**Design & CAD**: Figma, Pixelmator Pro, AutoCAD, Autodesk Inventor, Fusion 360